Somrik Banerjee

me25mtech14015@iith.ac.in | banerjeesomrik17@gmail.com | +91 8961885623 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.45
Kendriya Vidyalaya Ballygunge	Class XII (Central Board of Secondary Education)	2021	94.20%
Kendriya Vidyalaya Ballygunge	Class X (Central Board of Secondary Education)	2019	95.80%

M.TECH THESIS

Fractional-Order Nonlocal Damage Modeling & ABAQUS UEL Implementation

- Developed **fractional-order nonlocal damage mechanics formulations** for brittle and composite materials and implemented **FORTRAN-based UEL/UMAT subroutines in ABAQUS**, improving **damage localization and reducing mesh dependency** in fracture simulations.

PROJECTS

AI-Assisted Inverse Design of Composite Laminates

- Automated **Python-ABAQUS FEA simulations** to generate composite laminate datasets and developed **ANN/Random Forest models** for buckling-performance prediction across varied design parameters.
- Applied **Genetic Algorithm optimization** to achieve a **12.5% reduction in laminate weight** while satisfying specified buckling-load constraints.

ML-based prediction of fracture toughness incorporating boundary effects

- Developed an **Abaqus-Python framework** to generate parametric SIF datasets and trained **ML models** for fracture prediction across varying geometries and loading conditions.
- Validated ML predictions against established analytical solutions for fracture mechanics parameters.

Vibrational Analysis for Predicitive Maintenance of Shafts

- Developed a **MATLAB-based Jeffcott rotor model** to analyze vibration response and predict critical speed under varying rotational speeds.
- Implemented **1× phasor analysis** for shaft unbalance detection using vibration amplitude and phase measurements.
- Determined **unbalance magnitude, correction mass, and placement angle** for rotor balancing and predictive maintenance.

Fixed-Wing Search & Rescue UAV | Smart India Hackathon 2023

- Designed a fixed-wing Search & Rescue UAV, performing aircraft sizing and **NACA 6412 aerodynamic analysis** to determine lift, drag, thrust, climb performance and a **425 W power requirement** with safety factor.
- **Developed SolidWorks CAD models of the wing and structural components**, including ribs and spars, and proposed an onboard **optical/thermal imaging + deep-learning pipeline** for casualty detection and localization.

SKILLS

FEA & CAD : ANSYS | Abaqus | COMSOL | HyperMesh | Solidworks

Programming : Python | MATLAB | Fortran | MySQL

Abaqus Subroutines : UEL | UMAT | VUEL | VUMAT

RELEVANT COURSES

Analysis and Design of Composite Structures | Finite Element Method | Fracture Mechanics | Machine Learning |

Advanced Mechanics of Solids | Advanced Finite Element Method | Vibration & Dynamics

INTERNSHIP

Research Intern | NIT Durgapur

- Performed **numerical modelling of interfacial fluid flow**, analysing thermocapillary effects and liquid-film deformation using governing equations and nondimensional analysis.
- Developed a **MATLAB-based numerical solver** using finite-difference methods to simulate the evolution of thin-film interfaces under localized thermal gradients.

Vocational Trainee | Gun and Shell Factory, A unit of Advanced Weapons and Equipment Ltd.

- Gained hands-on exposure to **CNC machining and conventional manufacturing operations**, including turning, drilling, boring, reaming, threading and grinding.
- Executed a **guide-rod machining task** using an auto-feed lathe, covering centering, rough/smooth turning and finishing operations.

CONFERENCE PUBLICATION

ENERGY AND EXERGY ANALYSIS OF VAPOUR COMPRESSION REFRIGERATION SYSTEM

WITH LOW-GWP BASED NANOREFRIGERANT

Investigated AlO/R600a nanorefrigerant for enhanced thermodynamic performance of vapour compression refrigeration systems under varying operating conditions. Demonstrated improved coefficient of performance and exergy efficiency, alongside reduced exergy destruction, compared with conventional R600a.

POSITION OF RESPONSIBILITY

Placement Coordinator - IIT Hyderabad

Coordinated placement activities and facilitated communication between recruiters, students, and the institute.